

When it breaks

Every error below was hit for real while building this course. Find your message, apply the fix, carry on. Most of these take under a minute once you know what you are looking at.

FIRST, THREE QUESTIONS

1. Did you run the **setup cell** this session? Colab forgets everything when it recycles.
2. Did you run the cells **in order**? A variable defined in cell 8 does not exist if you jumped to cell 12.
3. Is the error at the **top** or the **bottom** of the red text? The bottom line names the problem; the lines above tell you which cell.

1 Access and setup

Project 'projects/xxx' not found or deleted

Your project ID is wrong, or that project no longer exists.

```
ee.Initialize(project='ee-yourname') # must match your Cloud project ID exactly
```

Find the real ID at console.cloud.google.com/earth-engine. It is the **ID**, not the display name — they are often different, and the ID usually has extra characters.

Earth Engine client library not initialized **or** Please authorize access

You skipped the authentication step, or your session recycled. Re-run the setup cell:

```
ee.Authenticate()  
ee.Initialize(project='ee-yourname')
```

THIS IS NOT A BUG

You will do this **every session**, and again whenever Colab disconnects you for being idle. It is the single most common "my notebook stopped working" cause in this course.

Caller does not have required permission **or the sign-in page refuses you**

Almost always a **university or workplace Google account** whose administrator has blocked third-party API access. Use a **personal Gmail** instead. Register a fresh Cloud project on it and you will be running in five minutes.

Antigravity gave me code that fails, or looks wrong

Treat it exactly like any other AI assistant: copy the whole error, paste it back into the same Antigravity chat, and ask for a fix. Do not let Antigravity run the code itself — copy the corrected version into Colab and run it there, so you can actually see the map.

Antigravity's chat has gone quiet or refuses to answer

You have most likely hit its weekly rate limit on the free tier. This is expected under heavy use — switch to **Gemini inside Colab** for the rest of the session and come back to Antigravity later. Nothing about your GEE work is lost; only the assistant you are prompting changes.

I ran my notebook inside Antigravity and the map did not show

This course was built and tested against **Colab**, not against running notebooks inside Antigravity itself. Copy your code into a Colab notebook and run it there — that is the supported path, and it is the one every lab in this course assumes.

Quota exceeded / User memory limit exceeded / **computation times out**

You asked for too much. Three fixes, in order of how much they help:

- **Raise** `scale`. Area statistics at `scale=100` are just as useful as at `scale=10` and cost a hundredth as much.
- **Shrink the area**. One province, not one country.
- **Add** `maxPixels=1e9, bestEffort=True` to every `reduceRegion`.

YOUR MONTHLY BUDGET IS REAL

A free account gets **150 EECU-hours per month**. Asking for a whole country at 10 m resolution can consume a large share of that in a single cell. If you exhaust it, you wait until next month — so form the habit now.

2 Loading data

Asset 'NASA/ORNL/biomass_carbon_density/v1' is not an Image

The ID is right; the *type* is wrong. This asset is an `ImageCollection` that contains exactly one image. Nearly every AI assistant gets this wrong.

```
# wrong
ornl = ee.Image('NASA/ORNL/biomass_carbon_density/v1')

# right
ornl = ee.ImageCollection('NASA/ORNL/biomass_carbon_density/v1').first()
```

The general rule: if `ee.Image()` fails, try `ee.ImageCollection()`, and vice versa. To check what you actually have, print the band names.

Image.load: Asset '...' not found

The dataset ID does not exist. Either a typo, or the assistant invented it. Check it against the cheat sheet. The most common real-world version of this is an **out-of-date Hansen version**:

```
# superseded – do not use
'UMD/hansen/global_forest_change_2024_v1_12'

# current as of August 2026
'UMD/hansen/global_forest_change_2025_v1_13'
```

Your collection has zero images

Nothing errored, but every result is empty or masked. Diagnose before you fix:

```
print('scenes:', collection.size().getInfo())
```

If that prints `0`, one of three things is wrong: your **dates** (is the satellite even flying then? Sentinel-2 starts in 2015), your **bounds** (is the AOI where you think it is?), or your **cloud filter** is too strict for a monsoon country.

```
Image.select: Pattern 'NDVI' did not match any bands
```

You asked for a band that does not exist. Indices like NDVI are **computed**, not supplied. See what you actually have:

```
print(img.bandNames().getInfo())
```

3 Numbers that are wrong but do not error

These are the dangerous ones. The code runs, the map looks lovely, and the answer is nonsense.

SYMPTOM	CAUSE	FIX
NDVI values in the thousands	Sentinel-2 values are stored *10000	<code>img.divide(10000)</code> before computing the index
Temperature around 300	ERA5 reports Kelvin	<code>.subtract(273.15)</code>
Rainfall far too small	You averaged daily values instead of summing them	<code>.sum()</code> over the month, then <code>reduceRegion</code> with a mean
Areas 10,000* too large	<code>pixelArea()</code> returns square metres	<code>.divide(1e4)</code> for hectares
Carbon exactly half what you expected	You applied *0.47 to ORNL, which is already carbon	Remove it. Apply 0.47 only to GEDI <code>MU</code> .
Classification accuracy 0.99+	Test points leaked into training	Split with <code>randomColumn</code> and score on the held-out set only
dNBR says half the province burned	Deciduous leaf-off looks like burn scar	Mask dNBR to pixels MCD64A1 confirms burned
Total area exceeds the province	You forgot to <code>.clip(aoi)</code> , or the reducer region is wrong	Clip the image and pass <code>geometry=aoi</code>

4 Colab and Python

```
NameError: name 'ee' is not defined
```

You ran a cell before the setup cell. Run cells top to bottom. If you have been jumping around, use **Runtime → Restart and run all** and let it go from the beginning.

```
NameError: name 'aoi' is not defined
```

Same cause, one cell later. The AOI cell has not run yet.

```
ModuleNotFoundError: No module named 'geemap'
```

The install did not run, or the runtime restarted since. Re-run:

```
!pip install -q geemap
```

AttributeError: 'Text' object has no property 'letterspacing'

A matplotlib property that does not exist — a classic AI invention. Matplotlib has no letter spacing option. Delete the argument. The general fix for any `... object has no property ...`: paste the error back and say *"that argument does not exist, please rewrite without it."*

The map cell shows nothing

The `Map` object has to be the **last line** of the cell, on its own:

```
Map.addLayer(image, vis, 'My layer')  
Map                                     # <- must be last, alone, no print()
```

Everything was working and now nothing does

Your Colab session recycled. All variables are gone and you are signed out of Earth Engine. **Runtime → Restart and run all**, then re-authenticate. Nothing is broken; you just have a fresh machine.

5 Still stuck after ten minutes

1. **Restart and run all.** This fixes a surprising share of problems caused by cells run out of order.
2. **Open the instructor notebook** for this lab and run its version of the cell. If that works and yours does not, compare the two directly.
3. **Give your assistant more context.** Paste the error *and* the cell that produced it *and* the cell that defines the variables it uses.
4. **Ask a neighbour.** Reading someone else's error out loud finds it faster than staring at your own.

KEEP THIS IN PERSPECTIVE

Hitting errors is not evidence that you are bad at this — it is what the work consists of. The people who look fluent are not making fewer mistakes; they are recognising them faster. Every error in this handout was one somebody made while writing the course.